

11.5 Fairness in Apportionment Methods

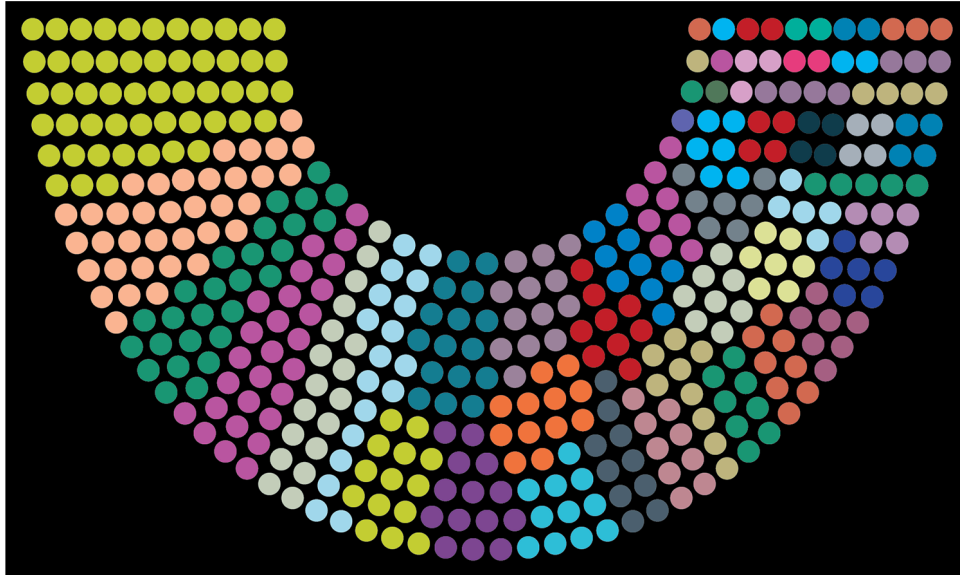


Figure 11.18 In this seating chart for the House of Representatives, each color indicates representatives from a particular state.

Learning Objectives

After completing this section, you should be able to:

1. Describe and illustrate the Alabama paradox.
2. Describe and illustrate the population paradox.
3. Describe and illustrate the new-states paradox.
4. Identify ways to promote fairness in apportionment methods.

Apportionment Paradoxes

The citizens of Imaginaria will want the apportionment method to be as fair as possible. There are certain characteristics that they would reasonably expect from a fair apportionment.

- If the house size is increased, the state quotas should all increase or remain the same but never decrease.
- If one state's population is growing more rapidly than another state's population, the faster growing state should not lose a seat while a slower growing state maintains or gains a seat.
- If there is a fixed number of seats, adding a new state should not cause an existing state to gain seats while others lose them.

However, apportionment methods are known to contradict these expectations. Before you decide on the right apportionment for Imaginarians, let's explore the **apportionment paradox**, a situation that occurs when an apportionment method produces results that seem to contradict reasonable expectations of fairness.

There is a lot that the founders of Imaginaria can learn from U.S. history. The constitution of the United States requires that the seats in the House of Representatives be apportioned according to the results of the census that occurs every decade, but the number of seats and the apportionment method is not stipulated. Over the years, several different apportionment methods and house sizes have been used and scrutinized for fairness. This scrutiny has led to the discovery of several of these apportionment paradoxes.

The Alabama Paradox

At the time of the 1880 U.S. Census, the Hamilton method of apportionment had replaced the Jefferson method. The number of seats in the House of Representatives was not fixed. To achieve the fairest apportionment possible, the house sizes were chosen so that the Hamilton and Webster methods would result in the same apportionment. The chief clerk of the Census Bureau calculated the apportionments for house sizes between 275 and 350. There was a surprising result that became known as the **Alabama paradox**, which is said to occur when an increase in house size reduces a state's quota. Alabama would receive eight seats with a house size of 299, but only receive seven seats if the house size increased to 300. (Michael J. Caulfield (Gannon University), "Apportioning Representatives in the United States Congress - Paradoxes of Apportionment," *Convergence* (November 2010), DOI:10.4169/loci003163)

EXAMPLE 11.42**The 1880 Alabama Quota**

The 1880 census recorded the population of Alabama as 1,513,401 and that of the U.S. as 62,979,766.

1. Calculate the standard divisor and standard quota for the State of Alabama based on a house size of 299.
2. Calculate the standard divisor and standard quota for the State of Alabama based on a house size of 300.
3. Did the standard quota increase or decrease when the house size increased?
4. Consider the Hamilton method of apportionment. Explain how Alabama's final quota could be smaller with a larger standard quota.

 Solution

1. The standard divisor is $62,979,766 \div 299 = 210$, resulting in 634.6689 citizens per seat. The standard quota for Alabama is $1,513,401 \div 210, 634.7 = 7.1850$ seats.
2. The standard divisor is $62,979,766 \div 300 = 209$, resulting in 932.5533 citizens per seat. The standard quota for Alabama is $1,513,401 \div 209, 932.6 = 7.2090$ seats.
3. The standard quota increased.
4. In each case, the state would receive the lower quota of 7 and then be awarded one more seat if the fractional part of the standard quota were high enough relative to the fractional parts of the other states' standard quotas. When the house size was 299, Alabama received one of the remaining seats after the lower quotas were distributed. When the house size was 300, Alabama did not receive one of the remaining seats after the lower quotas were distributed. It must have been the case that either the fractional part 0.2090 ranked lower amongst the other fractional parts of the state quotas than the fractional part 0.1850 did, or there were fewer remaining seats, or both.

After the 1900 census, the Census Bureau again calculated the apportionment based on various house sizes. It was determined that Colorado would receive three seats with a house size of 356, but only two seats with a house size of 357.

 YOUR TURN 11.42

The 1900 census recorded the population of Colorado as 539,700 and that of the U.S. as 76,212,168.

1. Calculate the standard divisor and standard quota for the State of Colorado based on a house size of 356.
2. Calculate the standard divisor and standard quota for the State of Colorado based on a house size of 357.
3. Did the standard quota increase or decrease when the house size increased?
4. Consider the Hamilton method of apportionment. Explain how Colorado's final quota could be smaller with a larger standard quota.

EXAMPLE 11.43**Hamilton's Method and the Alabama Paradox**

Suppose that States A and B each have a population of 6, while State C has a population of 2.

1. Use the Hamilton method to apportion 10 seats.
2. Use the Hamilton method to apportion 11 seats.
3. Does this example demonstrate the Alabama paradox? If so, how?

 Solution

1. **Step 1:** The total population is 14. The standard divisor is $14 \div 10 = 1.4$ individuals per seat.

Step 2: The states' standard quotas are as follows: $A \ 6 \div 1.4 \approx 4.29$, $B \ 6 \div 1.4 \approx 4.29$, and $C \ 2 \div 1.4 \approx 1.43$.

Step 3: The states' lower quotas are as follows: A 4, B 4, and C 1.

Step 4: The sum of the lower quotas is 9, which means there is one seat remaining to be apportioned. State C has the highest fractional part and receives the additional seat.

Step 5: The final apportionment is as follows: A 4, B 4, and C 2, which sums to 10.

2. **Step 1:** The total population is 14. The standard divisor is $14 \div 11 \approx 1.2727$ individuals per seat.

Step 2: The states' standard quotas are: $A \ 6 \div 1.2727 \approx 4.71$, $B \ 6 \div 1.2727 \approx 4.71$, and $C \ 2 \div 1.2727 \approx 1.57$.

Step 3: The states' lower quotas are: A 4, B 4, and C 1.

Step 4: The sum of the lower quotas is 9, which means there are two seats remaining to be apportioned. A and B have the highest fractional parts and receive the additional seats.

Step 5: The final apportionment is: A 5, B 5, and C 1.

3. Yes, this demonstrates the Alabama paradox because State C receives two seats if the house size is 10, but only one seat if the house size is 11.

YOUR TURN 11.43

Suppose that the founders of Imaginaria decide to have a parliament that apportions seats to four political parties based on the portion of the vote each party has earned. Also, suppose that Party A has 56.7 thousand votes, Party B has 38.5 thousand votes, Party C has 4.2 thousand votes, and Party D has 0.6 thousand votes.

1. Use the Hamilton method to apportion 323 seats.
2. Use the Hamilton method to apportion 324 seats.
3. Does this example demonstrate the Alabama paradox? If so, how?

The Population Paradox

It is important for the founders of Imaginaria to keep in mind that the populations of states change as time passes. Some populations grow and some shrink. Some populations increase by a large amount while others increase by a small amount. These changes may necessitate a **reapportionment** of seats, or the recalculation of state quotas due to a change in population. It would be reasonable for Imaginarians to expect that the state with a population that has grown more than others will gain a seat before the other states. Once again, this is not always the case with the Hamilton method of apportionment. The **population growth rate** of a state is the ratio of the change in the population to the original size of the population, often expressed as a percentage. This value is positive if the population is increasing and negative if the population is decreasing. The **population paradox** occurs when a state with an increasing population loses a seat while a state with a decreasing population either retains or gains seats. More generally, the population paradox occurs when a state with a higher population growth rate loses seats while a state with a lower population growth rate retains or gains seats.

Notice that the population paradox definition has two parts. If either part applies, then the population paradox has occurred. The first part of the definition only applies when one state has a decreasing population. The second part of the definition applies in all situations, whether there is a state with a decreasing population or not. It will be easier to identify situations that involve a decreasing population. The other situation requires the calculation of a growth rate. The reason that we don't have to calculate a growth rate when one state has a decreasing population and the other has an increasing population is that increasing population has a positive growth rate which is always greater than the negative growth rate of a decreasing population.

 *A state must lose a seat in order for the population paradox to apply. It is not enough for a state with a lower growth rate to gain a seat while a state with a higher growth rate retains the same number of seats.*

EXAMPLE 11.44

Apportionment of Respirators to Hospitals

Suppose that 18 respirators are to be apportioned to three hospitals based on their capacities. The Hamilton method is used to allocate the respirators in 2020, then to reallocate based on new capacities in 2021. The results are shown in the table below. How does this demonstrate the population paradox?

Hospital	Capacity in 2020	Respirators in 2020	Change in Capacity	Growth Rate = $\frac{\text{Change in Capacity}}{\text{Capacity in 2020}}$	Capacity in 2021	Respirators in 2021
A	825	9	57	$\frac{57}{825} \approx 0.0691 = 6.91\%$	882	9
B	613	7	13	$\frac{13}{613} \approx 0.0212 = 2.12\%$	626	6
C	239	2	3	$\frac{3}{239} = 0.0126 = 1.26\%$	242	3

 Solution

Hospital B lost a respirator while hospital C gained one, even though hospital B had a higher growth rate than hospital C.

 YOUR TURN 11.44

1. Suppose that 18 respirators are to be apportioned to three hospitals based on their capacities. The respirators are allocated based on the Hamilton method in 2020, then reallocated based on new capacities in 2021. The results are shown in the table below. How does this demonstrate the population paradox?

Hospital	Capacity in 2020	Respirators in 2020	Change in Capacity	Growth Rate = $\frac{\text{Change in Capacity}}{\text{Capacity in 2020}}$	Capacity in 2021	Respirators in 2021
A	237	5	6	$6 \div 237 = 0.0253 = 2.53\%$	243	6
B	889	21	69	$69 \div 889 \approx 0.0776 = 7.76\%$	958	21
C	674	16	18	$18 \div 674 \approx 0.0267 = 2.67\%$	692	15

FORMULA

The growth rate of a population can be calculated by subtracting the previous population size from the current population size, and then dividing the difference by the previous population size.

$$\text{population growth rate} = \frac{\text{current population size} - \text{previous population size}}{\text{previous population size}}$$

 Make sure to calculate the subtraction before the division. If you are entering the values in a calculator, it is helpful to put parentheses around the subtracted terms.

EXAMPLE 11.45

The Congress of Costaguana

The country of Costaguana has three states: Azuera with a population of 894,000; Punta Mala with a population of 696,000; and Esmeralda with a population of 215,000. There are 38 seats in the Congress of Costaguana. The apportionment of the seats is determined by Hamilton's method to be: 19 for Azuera, 15 for Punta Mala, and 4 for Esmeralda. A census reveals that the population has grown and the seats must be reapportioned. If Azuera now has 953,000 residents, Punta Mala now has 706,000 residents, and Esmeralda now has 218,000 residents, how many seats

will each state receive upon reapportionment? How is this an example of the population paradox?

 Solution

The Hamilton reapportionment is: 19 for Azuera, 14 for Punta Mala, and 5 for Esmerelda. This is an example of the population paradox because Punta Mala lost a seat to Esmerelda, even though Punta Mala's population grew by 1.44 percent while Esmerelda's only grew by 1.40 percent.

 YOUR TURN 11.45

1. The country of Elbonia has three states: Mudston with a population of 866,000; WallaWalla with a population of 626,000; and Dilberta with a population of 256,000. There are 38 seats in the Congress of Elbonia. The apportionment of the seats is determined by Hamilton's method to be: 19 for Mudston, 14 for WallaWalla, and 5 for Dilberta. A census reveals that the population has grown and the seats must be reapportioned. If Mudston now has 921,000 residents, WallaWalla now has 640,000 residents, and Dilberta now has 260,000 residents, how many seats will each state receive upon reapportionment? How is this an example of the population paradox?

The New-States Paradox

As a founder of Imaginaria, you might also consider the possibility that Imaginaria could annex nearby lands and increase the number of states. This occurred several times in the United States such as when Oklahoma became a state in 1907. The House size was increased from 386 to 391 to accommodate Oklahoma's quota of five seats. When the seats were reapportioned using Hamilton's method, New York lost a seat to Maine despite the fact that their populations had not changed. This is an example of the **new-state paradox**, which occurs when the addition of a new state is accompanied by an increase in seats to maintain the standard ratio of population to seats, but one of the existing states loses a seat in the resulting reapportionment.

EXAMPLE 11.46

New State of Oscuridad

The country of San Lorenzo has grown from two states to three. The house size of the congress has been increased by eight and the seats have been reapportioned to accommodate the new state of Oscuridad. The constitution mandates the use of the Hamilton method of apportionment. Use this information and the following table to answer the questions.

State	Population (in hundreds)	Original Apportionment	Reapportionment
Clara	7,100	39	40
Velasco	9,080	51	50
Oscuridad	1,500	Not Applicable	8

1. What was the original house size?
2. What is the new house size?
3. How is this reapportionment an example of the new-states paradox?

 Solution

1. $39 + 51 = 90$
2. $40 + 50 + 8 = 98$
3. The original state of Velasco lost a seat to the original state of Clara when the new state of Oscuridad was added.

 YOUR TURN 11.46

The country of Narnia has grown from two states to three. The house size of the congress has been increased by five

and the seats have been reapportioned to accommodate the new state of Chippingford. The constitution mandates the use of the Hamilton method of apportionment. Use this information and the following table to answer the questions.

State	Population (in hundreds)	Original Apportionment	Reapportionment
Beruna	7600	39	40
Beaversdam	9720	51	50
Chippingford	1000	Not Applicable	5

1. What was the original house size?
2. What is the new house size?
3. How is this reapportionment an example of the new-states paradox?

EXAMPLE 11.47

The Growing Country of Gulliversia

The country of Gulliversia has two states: Lilliput with a population of 700,000 and Brobdingnag with a population of 937,000. The constitution of Gulliversia requires that the 90 congressional seats be apportioned by Hamilton's method. Lilliput has received 38 seats while Brobdingnag has received 52 seats. Recently, the island of Houyhnhnmsland with a population of 170,000 has joined the union, becoming a state of Gulliversia. When Houyhnhnmsland is included, nine additional seats must be apportioned to maintain the same ratio of seats to citizens. Use Hamilton's method to reapportion the 99 seats to the three states. How is the resulting apportionment an example of the new-states paradox?

Solution

The reapportionment gives 39 seats to Lilliput, 51 seats to Brobdingnag, and 9 seats to Houyhnhnmsland. This is an example of the new-states paradox because the original state of Brobdingnag lost a seat to the original state of Lilliput when the new state was added to the union.

YOUR TURN 11.47

1. The country of Neverland has two states: Neverwood with a population of 760,000 and Mermaids Lagoon with a population of 943,000. The constitution of Neverland requires that the 84 congressional seats be apportioned by Hamilton's method. Neverwood has received 37 seats while Mermaids Lagoon has received 47 seats. Recently, the island of Marooners Rock with a population of 190,000 has joined the union, becoming a state of Neverland. When Marooners Rock is included, nine additional seats must be apportioned to maintain the same ratio of seats to citizens. Use Hamilton's method to reapportion the 93 seats to the three states. How is the resulting apportionment an example of the new-states paradox?

When a new state is added, it is necessary to determine the amount that the house size must be increased to retain the original ratio of population to seats, in other words to keep the original standard divisor. To calculate the new house size, divide the new population by the original standard divisor, and round to the nearest whole number.

FORMULA

$$\text{New House Size} = \frac{\text{New Population}}{\text{Original Standard Divisor}} \text{ rounded to the nearest whole number.}$$

EXAMPLE 11.48**Oklahoma Joins the Union**

Oklahoma was admitted as the 46th state on November 16, 1907. Before Oklahoma joined the union, the U.S. population was approximately 75,030,000 and the House of Representatives had 386 seats. The new state had a population of approximately 970,000. Use this information to estimate the original standard divisor to the nearest hundred, the new population, the new house size, and the number of seats Oklahoma should receive.

 Solution

Step 1: Original Standard Divisor $\approx \frac{75,030,000}{386} \approx 194,400$

Step 2: New Population $\approx 75,030,000 + 970,000 = 76,000,000$

Step 3: New House Size $\approx \frac{76,000,000}{194,400} \approx 391$

Step 4: There are $391 - 386 = 5$ new seats to be apportioned to Oklahoma.

 YOUR TURN 11.48

1. New Mexico was admitted as the 47th state on January 6, 1912. Before New Mexico joined the union, the U.S. population was approximately 76,000,000 and the House of Representatives had 391 seats. The new state had a population of approximately 300,000. Use this information to estimate the original standard divisor to the nearest hundred, the new population, the new house size, and the number of seats New Mexico should receive.

The Search for the Perfect Apportionment Method

The ideal apportionment method would simultaneously satisfy the following four fairness criteria.

- Satisfy the quota rule
- Avoid the Alabama paradox
- Avoid the population paradox
- Avoid the new-states paradox

We have seen that the Hamilton method allows the Alabama paradox, the population paradox, and the new-states paradox in some apportionment scenarios. Let's explore the results of the other methods of apportionment we have discussed in some of the same scenarios.

EXAMPLE 11.49**Orange Grove and the New-States Paradox**

The incorporated town of Orange Grove consists of two subdivisions: The Oaks with 1,254 residents and The Villages with 10,746 residents. A council with 100 members supervises the municipality's operations. The council votes to annex an unincorporated subdivision called The Lakes with a population of 630. They plan to increase the size of the council to maintain the ratio of seats to residents such that the new council will have 100 seats plus the number of seats given to The Lakes. Use each of the following apportionment methods and indicated number of additional seats to find the original and new apportionment and determine whether the new-state paradox occurs.

1. Jefferson's method with five additional seats.
2. Adams's method with six additional seats.
3. Webster's method with five additional seats

 Solution

1. Using a modified divisor of 119, the original apportionment would have been: The Oaks 10 and The Villages 90. Using a modified divisor of 119, the new apportionment would be: The Oaks 10, The Villages 90, and The Lakes 5. The new-state paradox does not occur.
2. Using a modified divisor of 121, the original apportionment would have been: The Oaks 11 and The Villages 89. Using a modified divisor of 121, the new apportionment would be: The Oaks 11, The Villages 89, and The Lakes 6. The new-state paradox does not occur.

3. Using the standard divisor of 120, the original apportionment would have been: The Oaks 10 and The Villages 90. Using a modified divisor of 119.5, the new apportionment would be: The Oaks 10, The Villages 90, and The Lakes 5. The new-state paradox does not occur.

YOUR TURN 11.49

1. Suppose that in 2016, States A, B, and C had populations of 13 million, 12 million, and 112 million, respectively. In 2020, State A has grown by 1 million residents, State B has lost 1 million residents, and State C has gained 2 million residents. Compare the apportionments in 2016 to 2020 using each method given below. Which of the four methods violate(s) the population paradox in this scenario?

We have seen in our examples that neither the population paradox nor the new-states paradox occurred when using the Jefferson, Adams, and Webster methods. It turns out that, although all three of these divisor methods violate the quota rule, none of them ever causes the population paradox, new-states paradox, or even the Alabama paradox. On the other hand, the Hamilton method satisfies the quota rule, but will cause the population paradox, the new-states paradox, and the Alabama paradox in some scenarios.

In 1983, mathematicians Michel Balinski and Peyton Young proved that no method of apportionment can simultaneously satisfy all four fairness criteria.

There are other apportionment methods that satisfy different subsets of these fairness criteria. For example, the mathematicians, Balinski and Young who proved the Balinski-Young Impossibility Theorem created a method that both satisfies the quota rule and is free of the Alabama paradox. (Balinski, Michel L.; Young, H. Peyton (November 1974). "A New Method for Congressional Apportionment." *Proceedings of the National Academy of Sciences*. 71 (11): 4602–4606.) However, no method may always follow the quota rule and simultaneously be free of the population paradox. (Balinski, Michel L.; Young, H. Peyton (September 1980). "The Theory of Apportionment" (PDF). Working Papers. International Institute for Applied Systems Analysis. WP-80-131.)

So, as you and your fellow founders of Imaginaria make the important decision about the right apportionment method for Imaginaria, do not look for a perfect apportionment method. Instead, look for an apportionment method that best meets the needs and concerns of Imaginarians.

Check Your Understanding

In the following questions, a scenario is given. Determine whether the scenario is an example of a quota rule violation, the Alabama paradox, the population paradox, the new-states paradox, or none of these.

33. A city purchased five new firetrucks and apportioned them among the existing fire stations. Although your neighborhood fire station has the same proportion of the city's firetrucks as before the new ones were purchased, it now has one fewer.
34. The school resources officers in a county were reapportioned based on the most recent census. The number of students at Chapel Run Elementary went up while the number of students at Panther Trail Elementary went down. However, Chapel Run now has one fewer resources officer while Panther Trail has one more than it did previously.
35. The standard quota for States A, B, and C are 1.19, 2.73, and 5.71 respectively. State A received 1 seat, State B received 3 seats, and State C received 4 seats.
36. A corporation that owns several hospitals purchased an additional hospital, causing the doctor to patient ratio to decrease.
37. When the city of Cocoa annexed an adjacent unincorporated community, the number of seats on the city council was increased to maintain the standard ratio of citizens to seats, but one existing community of Cocoa still lost a seat on the city council to another existing community of Cocoa when the new community was added.

In the following questions, a type of scenario is described. Indicate which paradox could arise in a scenario of this kind.

38. A reapportionment occurs because the populations of the states change, and the house size remains the same.
39. A reapportionment occurs because the house size increases, and the populations of the states remain the same.
40. A reapportionment occurs because an additional state is added to the union, the populations of the original states remain the same, and the house size is increased to correspond to the population of the new state.



SECTION 11.5 EXERCISES

In the following exercises, determine whether the scenario violates the Alabama paradox. Justify your answer.

1. A company with an office in each of four cities must distribute 145 new Chromebooks to the four offices. It is determined that Office A will receive 42, Office B will receive 17, Office C will receive 35, and Office D will receive 51. At the last minute, it is discovered that there are 146 Chromebooks. When they are reapportioned, Office A receives 42, Office B receives 16, Office C receives 36, and Office D receives 52.
2. A county with three towns has 30 garbage trucks to apportion. Attenborough receives 6 trucks, Breckenridge receives 8 trucks, and Cabbotsville receives 16 trucks, in proportion to their populations. When one additional truck is purchased, the reallocation results in 5 trucks for Attenborough, 9 for Breckenridge, and 8 for Cabbotsville though there has been no change in populations.
3. A group of 60 dentists work for a company that runs five offices. The dentists have been apportioned to the offices by the number of patients. Office A receives 14 dentists, Office B receives 11, Office C receives 11, Office D receives 12, and Office E receives 12. When a new dentist joins the group, the new apportionment gives the following: A 14, B 11, C 11, D 12, and E 13.

In the following exercises, determine whether the scenario violates the population paradox. Justify your answer.

4. A company with locations in three cities plans to give 200 achievement awards that shall be apportioned to the three cities by population. City A with 9,150 employees is allotted 61 awards, City B with 6,040 employees is allotted 40 awards, and City C with 14,810 employees is allotted 99 awards to distribute. Then it is discovered that the number of employees is out of date. The awards are reallocated based on the new populations: City A with 9,180; City B with 6,040; and City C with 14,930. It turns out that the apportionment remains the same.
5. A soccer club must apportion soccer balls to the teams among four age brackets based on the number of teams in each bracket. Bracket U8 receives 32 balls, U12 receives 46 balls, U15 receives 29 balls, and U18 receives 25 balls. Mid-season, the balls are reapportioned. U8 has decreased by 10 percent, U12 has increased by 20 percent, U15 remains the same, and U18 has increased by 10 percent. The reapportionment gives 33 balls to U8, 46 balls to U12, 29 balls to U15, and 14 balls to U18.
6. READ (Reading Education Assistance Dogs) trains and certifies therapy animals to serve in classrooms and help children develop a love of reading in a low-stress environment. Suppose that 20 therapy teams are apportioned to three schools based on their populations. When the population of School A decreases by 10 percent, the population of School B increases by 10 percent, and the population of School C remains constant, the apportionment remains the same.

In the following exercises, determine whether the scenario violates the new-states paradox. Justify your answer.

7. A charity organization has 851 volunteers in Country A and 3449 volunteers in Country B. There are 43 lead organizers that must be apportioned to the two locations. Country A receives 9 while Country B receives 34. When the operations are expanded to Country C with 725 volunteers, 7 new lead organizers are added to the team. When the lead organizers are reapportioned, Country A receives 9, Country B receives 34, and Country C receives 7 lead organizers.
8. A country has two states. There are 22 seats in the legislature. State A has 4 seats, and State B has 18 seats. When State C joins the union, the number of representatives is increased by five. Under the new apportionment, State A receives 3 seats, State B receives 19 seats, and State C receives 5 seats.
9. In the Garunga Solar System, there are four inhabited planets. Three of the planets are members of the United Association of Garungan Planets (UAGP). The UAGP has 67 seats. Planet Angluertka has 40 seats, planet Bangluertka has 27 seats, and planet Clangluertka has 13 seats. When planet Danggluertka decides to join the UAGP, 14 seats are added to accommodate them proportionately. The new allocation of seats gives 40 seats to Angluertka, 26 seats to Bangluertka, 14 seats to Clangluertka, and 14 seats to Danggluertka.

In the following exercises, use the Hamilton method of apportionment to answer the questions.

10. When the number of seats changed from 147 to 148, the standard quotas changed from A 44.24, B 17.35, C 37.12, and D 48.29 to A 45.54, B 17.47, C 37.37, and D 48.62.
 - a. How did the increase in seats impact the apportionment?
 - b. Is this apportionment an example of a paradox? Justify your answer.
11. When the number of seats changed from 126 to 127, the standard quotas changed from A 9.57, B 29.49, C 33.89, D 28.43, and E 24.61 to A 9.65, B 29.72, C 34.16, D 38.66, and E 24.81.
 - a. How did the increase in seats impact the apportionment?
 - b. Is this apportionment an example of a paradox? Justify your answer.
12. When the number of seats changed from 25 to 26, the standard quotas changed from A 2.21, B 5.25, C 11.27,

- and D 6.27 to A 2.30, B 5.46, C 11.72, and D 6.52.
- How did the increase in seats impact the apportionment?
 - Is this apportionment an example of a paradox? Justify your answer.
- When the number of seats changed from 25 to 26, the standard quotas changed from A 2.43, B 5.42, and C 8.15 to A 2.46, B 5.47, and C 8.07.
 - How did the increase in seats impact the apportionment?
 - Is this apportionment an example of a paradox? Justify your answer.
 - The house size is 18. When the population of State A increases by 11.76 percent, State B increases by 16.22 percent, and State C increases by 12.18 percent, the standard quotas change from A 2.46, B 6.00, and C 9.53 to A 2.41, B 6.25, and C 9.34.
 - How did the change in populations impact the apportionment?
 - Is this apportionment an example of a paradox? Justify your answer.
 - The house size is 100. When the population of State A increases by 20 percent, State B increases by 10 percent, State C increases by 30 percent, and the populations of States D, E, and F remain the same. The standard quotas change from A 12.50, B 25.00, C 9.38, D 18.75, E 12.50 and F 21.88 to A 13.91, B 25.51, C 11.30, D 17.39, E 11.59, and F 20.29.
 - How did the change in populations impact the apportionment?
 - Is this apportionment an example of a paradox? Justify your answer.
 - The house size is 17. When the population of State A increases by 12.73 percent, State B increases by 15.63 percent, and State C increases by 12.90 percent, the standard quotas change from A 2.53, B 5.90, and C 8.57 to A 2.51, B 5.99, and C 8.50.
 - How did the change in populations impact the apportionment?
 - Is this apportionment an example of a paradox? Justify your answer.
 - The house size is 38 seats. When the population of State A increases by 6.60 percent, State B increases by 1.44 percent, and State C increases by 1.40 percent, the standard quotas change from A 18.82, B 14.65, and C 4.53 to A 19.59, B 14.29, and C 4.41.
 - How did the change in populations impact the apportionment?
 - Is this apportionment an example of a paradox? Justify your answer.
 - The house size is 24 seats. When the population of State A increases by 28 percent, State B increases by 26 percent, and State C increases by 15 percent, the standard quotas change from A 3.38, B 6.32, and C 14.30 to A 3.63, B 6.67, and C 13.71.
 - How did the change in populations impact the apportionment?
 - Is this apportionment an example of a paradox? Justify your answer.
 - The house size was 60. There were three states with standard quotas of A 4.18, B 15.38, and C 40.44. A fourth state was annexed, and the house size was increased to 65. The new standard quotas are A 4.17, B 15.33, C 40.32, and D 5.18.
 - How did the additional state impact the apportionment?
 - Is this apportionment an example of a paradox? Justify your answer.
 - The house size was 50. There were three states with standard quotas of A 9.41, B 24.42, and C 16.17. A fourth state was annexed, and the house size was increased to 66. The new standard quotas are A 9.36, B 24.30, C 16.09, and D 16.25.
 - How did the additional state impact the apportionment?
 - Is this apportionment an example of a paradox? Justify your answer.
 - The house size was 27. There were three states with standard quotas of A 6.39, B 11.40, and C 9.21. A fourth state was annexed, and the house size was increased to 35. The new standard quotas are A 6.38, B 11.37, C 9.19, and D 8.06.
 - How did the additional state impact the apportionment?
 - Is this apportionment an example of a paradox? Justify your answer.
 - The house size was 100. There were three states with standard quotas of A 26.09, B 30.43, and C 43.48. A fourth state was annexed, and the house size was increased to 122. The new standard quotas are A 26.14, B 30.50, C 43.57, and D 21.78.
 - How did the additional state impact the apportionment?
 - Is this apportionment an example of a paradox? Justify your answer.

In the following exercises, use the information in the table below.

State	A	B	C	D	E	F	G	H	I	J	K	L	P	Q	R
Population	624	1,219	979	3,462	7,470	4,264	5,300	263	809	931	781	676	150	250	350
Original House Size	38			204				126					50		
Updated House Size	39			205				127					51		

23. Consider States A, B, and C.
 - a. Determine the apportionment for States A, B, and C with the original house size using the Hamilton method.
 - b. Determine the apportionment for States A, B, and C with the updated house size using the Hamilton method.
 - c. Does the change in the house size and use of the Hamilton method cause the Alabama paradox? Explain your reasoning.
24. Consider States D, E, F, and G.
 - a. Determine the apportionment for States D, E, F, and G with the original house size using the Hamilton method.
 - b. Determine the apportionment for States D, E, F, and G with the updated house size using the Hamilton method.
 - c. Does the change in the house size and use of the Hamilton method cause the Alabama paradox? Explain your reasoning.
25. Consider States H, I, J, K, and L.
 - a. Determine the apportionment for States H, I, J, K, and L with the original house size using the Hamilton method.
 - b. Determine the apportionment for States H, I, J, K, and L with the updated house size using the Hamilton method.
 - c. Does the change in the house size and use of the Hamilton method cause the Alabama paradox? Explain your reasoning.
26. Consider States P, Q, and R.
 - a. Determine the apportionment for States P, Q, and R with the original house size using the Hamilton method.
 - b. Determine the apportionment for States P, Q, and R with the updated house size using the Hamilton method.
 - c. Does the change in the house size and use of the Hamilton method cause the Alabama paradox? Explain your reasoning.

In the following exercises, use the information in the table below.

State	Original Population	Updated Population	Population Growth Rate	House Size
A	889	958	7.76%	
B	674	692	2.67%	42
C	237	243	2.53%	
D	12,032	14,124	$d\%$	
E	10,789	9,726	$e\%$	135

State	Original Population	Updated Population	Population Growth Rate	House Size
F	995	2,304	$f\%$	
G	901	1,156	28.3%	
H	1,683	2,125	26.3%	24
I	3,808	4,369	14.7%	
X	56	63	$x\%$	
Y	125	141	$y\%$	16
Z	182	213	$z\%$	
P	6,534	6,534	$p\%$	
Q	7,832	7,810	$q\%$	40
R	13,959	13,992	$r\%$	
S	20,515	21,164	$s\%$	

27. Calculate the population growth rates d , e , and f for States D, E, and F. Give answer as a percentage rounded to one decimal place.
28. Calculate the population growth rates p , q , r , and s for States P, Q, R, and S. Give answer as a percentage rounded to one decimal place.
29. Calculate the population growth rates x , y , and z for States X, Y, and Z. Give answer as a percentage rounded to one decimal place.
30. Consider States A, B, and C.
 - a. Determine the Hamilton apportionment for States A, B, and C with the original population.
 - b. Determine the Hamilton apportionment for States A, B, and C with the updated population.
 - c. Does the increase in population of States A, B, and C from the original population to the updated population and the use of the Hamilton method cause the population paradox? Explain your reasoning.
31. Consider States D, E, and F.
 - a. Determine the Hamilton apportionment for States D, E, and F with the original population.
 - b. Determine the Hamilton apportionment for States D, E, and F with the updated population.
 - c. Does the increase in population and the use of the Hamilton method cause the population paradox? Explain your reasoning.
32. Consider States G, H, and I.
 - a. Determine the Hamilton apportionment for States G, H, and I with the original population.
 - b. Determine the Hamilton apportionment for States G, H, and I with the updated population.
 - c. Does the increase in population and the use of the Hamilton method cause the population paradox? Explain your reasoning.
33. Consider States X, Y, and Z.
 - a. Determine the Hamilton apportionment for States X, Y, and Z with the original population.
 - b. Determine the Hamilton apportionment for States X, Y, and Z with the updated population.
 - c. Does the increase in population and the use of the Hamilton method cause the population paradox? Explain your reasoning.
34. Consider States P, Q, R, and S.
 - a. Determine the Hamilton apportionment for States P, Q, R, and S with the original population.
 - b. Determine the Hamilton apportionment for States P, Q, R, and S with the updated population.

- c. Does the increase in population and the use of the Hamilton method cause the population paradox? Explain your reasoning.

In the following exercises, use the information in the table below.

State	Population	Original House Size	New House Size
A	627		
B	1,287	25	
C	973		32
D	815		
E	520		
F	1,510	50	
G	1,060		h
H	950		
P	1,222	100	
Q	473		
R	225		r
K	1,688		
L	7,912	48	
M	1,448		m
T	150		
U	250		
V	350	50	w
W	450		

35. Consider States A, B, and C.
- Calculate the standard divisor based on the original house size.
 - Use the Hamilton method to apportion the seats.
36. Consider States E, F, and G.
- Calculate the standard divisor based on the original house size.
 - Use the Hamilton method to apportion the seats.
37. Consider States P and Q.
- Calculate the standard divisor based on the original house size.
 - Use the Hamilton method to apportion the seats.
38. Consider States K and L.

- a. Calculate the standard divisor based on the original house size.
 - b. Use the Hamilton method to apportion the seats.
39. Suppose that States A, B, and C annex State D and increase the house size proportionately.
 - a. Calculate the standard divisor based on the new house size.
 - b. Use the Hamilton method to reapportion the seats.
 - c. Does the new-states paradox occur?
40. Suppose that States E, F, and G annex State H and increase the house size proportionately.
 - a. Determine the new house size, h , that is necessary.
 - b. Calculate the standard divisor based on the new house size.
 - c. Use the Hamilton method to reapportion the seats.
 - d. Does the new-states paradox occur?
41. Suppose that States P and Q annex State R and increase the house size proportionately.
 - a. Determine the new house size, r , that is necessary.
 - b. Calculate the standard divisor based on the new house size.
 - c. Use the Hamilton method to reapportion the seats.
 - d. Does the new-states paradox occur?
42. Suppose that States K and L annex State M and increase the house size proportionately.
 - a. Determine the new house size, m , that is necessary.
 - b. Calculate the standard divisor based on the new house size.
 - c. Use the Hamilton method to reapportion the seats.
 - d. Does the new-states paradox occur?
43. Suppose that States T, U, and V annex State W and increase the house size proportionately.
 - a. Calculate the standard divisor based on the original house size and only States T, U, and V.
 - b. Use the Hamilton method to apportion the seats to T, U, and V.
 - c. Determine the new house size when State W is annexed.
 - d. Calculate the standard divisor based on the new house size.
 - e. Use the Hamilton method to reapportion the seats.
 - f. Does the new-states paradox occur? (Refer to part b.)
44. Suppose 24 seats are apportioned to States A, B, and C with populations of 16, 15, and 125 respectively. Then the populations of States A, B, and C change to 17, 15, and 126 respectively.
 - a. Demonstrate that the population paradox occurs when the Hamilton method is used.
 - b. Determine whether the population paradox occurs when the Webster method is used. Justify your answer.
45. Suppose that 10 seats are apportioned to States A, B, and C with populations 6, 6, and 2 respectively. Then the number of seats is increased to 11. Demonstrate that the Alabama paradox occurs when the Hamilton method is used.